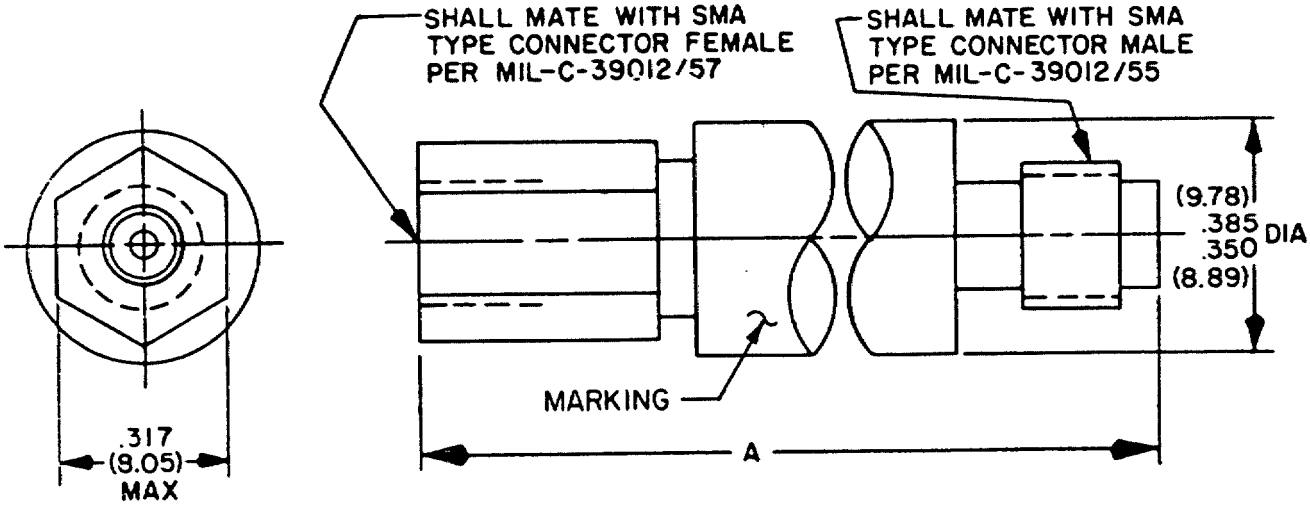
MIL-A-3933/14C
24 April 1985
SUPERSEDING
MIL-A-3933/14B
30 June 1981

MILITARY SPECIFICATION SHEET

ATTENUATORS, FIXED, COAXIAL LINE (SMA CONNECTORS), FREQUENCY RANGE: DC TO 12.4 GHz, CLASS IV, LOW POWER

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the attenuators described herein shall consist of this specification and the latest issue of MIL-A-3933.



NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Metric equivalents are in parentheses.

FIGURE 1. Dimensions and configuration.

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ENGINEERING DATA:

Class: IV, except for temperature sensitivity.

Operating frequency range: dc to 12.4 GHz.

Test frequencies for VSWR and attenuation measurements: dc, and 4, 8, and 12.4 GHz.

Basic test frequency: 8.0 GHz.

Operating temperature range: -55°C to +125°C.

Maximum weight: See table I.

REQUIREMENTS:

Dimensions and configuration: See figure 1.

Nominal impedance: 50 ohms.

Attenuation: See table I.

Power: See table I.

VSWR: See table I.

Temperature sensitivity: $\pm 0.0001 \frac{dB}{dB \times °C}$ maximum.

Part number: M3933/14- (and dash number from table I).

Supersession data: See table II.

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TABLE I. Electrical characteristics.

Dash number	Attenuation dB			Maximum power input 0 25°C		VSWR maximum 1/			Weight (oz) max	Dimension A (inches) maximum
	Nominal	Deviation		AV (W) 3/ (continuously)	Peak 2/ (kW)	dc to 4.0 GHz	4.0 to 8.0 GHz	8.0 to 12.4 GHz		
		DC	DC to 12.4 GHz							
01 N S	3	±0.2	±0.3	2	0.2	1.15:1	1.20:1	1.25:1	0.5	1.240 (31.50)
02 N S	6	±0.2	±0.3	2	0.2	1.15:1	1.20:1	1.25:1	0.5	1.240 (31.50)
03 N S	10	±0.2	±0.3	2	0.2	1.15:1	1.20:1	1.25:1	0.5	1.240 (31.50)
04 N S	20	±0.3	±0.5	2	0.2	1.15:1	1.20:1	1.25:1	0.5	1.240 (31.50)
05 N S	15	±0.3	±0.5	2	0.2	1.15:1	1.20:1	1.25:1	0.5	1.240 (31.50)
06 N S	1	±0.2	±0.3	2	0.2	1.15:1	1.20:1	1.25:1	0.5	1.240 (31.50)
07 N S	2	±0.2	±0.3	2	0.2	1.15:1	1.20:1	1.25:1	0.5	1.240 (31.50)
08 N S	4	±0.2	±0.3	2	0.2	1.15:1	1.20:1	1.25:1	0.5	1.240 (31.50)
09 N S	5	±0.2	±0.3	2	0.2	1.15:1	1.20:1	1.25:1	0.5	1.240 (31.50)
10 N S	7	±0.2	±0.4	2	0.2	1.15:1	1.20:1	1.25:1	0.5	1.240 (31.50)
11 N S	8	±0.2	±0.4	2	0.2	1.15:1	1.20:1	1.25:1	0.5	1.240 (31.50)
12 N S	9	±0.2	±0.4	2	0.2	1.15:1	1.20:1	1.25:1	0.5	1.240 (31.50)
13 N S	30	±0.4	±0.9	2	0.2	1.15:1	1.20:1	1.25:1	0.5	1.850 (46.99)

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TABLE I. Electrical characteristics - Continued.

Dash number	Attenuation dB			Maximum power input @ 25°C		VSWR maximum 1/			Weight (oz) max	Dimension A (inches) maximum
	Nominal	Deviation		AV (W) 3/ (continuously)	Peak 2/ (kW)	dc to 4.0 GHz	4.0 to 8.0 GHz	8.0 to 12.4 GHz		
		DC	DC to 12.4 GHz							
14 N S	40	±0.4	±0.9	2	0.2	1.15:1	1.20:1	1.25:1	0.5	1.850 (46.99)
15 N S	60	±1.0	±2.0	2	0.2	1.15:1	1.20:1	1.25:1	0.5	1.850 (46.99)
16 N S	50	±1.0	±2.0	2	0.2	1.15:1	1.20:1	1.25:1	0.5	1.850 (46.99)
17 N S	28	±0.4	±0.9	2	0.2	1.15:1	1.20:1	1.25:1	0.5	1.850 (46.99)
18 N S	16	±0.3	±0.5	2	0.2	1.15:1	1.20:1	1.25:1	0.5	1.240 (31.50)
19 N S	14	±0.3	±0.5	2	0.2	1.15:1	1.20:1	1.25:1	0.5	1.240 (31.50)
20 N S	13	±0.3	±0.5	2	0.2	1.15:1	1.20:1	1.25:1	0.5	1.240 (31.50)
21 N S	12	±0.3	±0.5	2	0.2	1.15:1	1.20:1	1.25:1	0.5	1.240 (31.50)
22 N S	11	±0.3	±0.5	2	0.2	1.15:1	1.20:1	1.25:1	0.5	1.240 (31.50)
23 N S	1.5	±0.2	±0.3	2	0.2	1.15:1	1.20:1	1.25:1	0.5	1.240 (31.50)
24 N S	31	±0.4	±0.9	2	0.2	1.15:1	1.20:1	1.25:1	0.5	1.850 (46.99)

1/ VSWR value is for both ends.

2/ Peak power for a duty cycle of 5×10^{-4}; maximum pulse duration 5 microseconds.

3/ Power input is derated linearly from +25°C to 0.5 watts at +125°C.

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TABLE II. Supersession data.

Part number M3933/14-	Supersedes M3933/15-
01	01
02	02
03	03
04	04
13	05
14	06
13	07
14	08

NOTE: Attenuators covered by this specification sheet are substitutable for the manufacturer's or user's part numbers specified in table III. This information in no way implies that the manufacturer's part number is suitable as a substitute for the military part number.

TABLE III. Substitution information.

Military part number M3933/14-	Manufacturer's code	Manufacturer's part number	NSN
01	99899	4778-3	5985-00-152-7923
01	16179	20610-3	5985-00-152-7923
01	49956	2888573-1	---
01	93459	3M-3	5985-00-152-7923
01	07397	26-807036-3	5985-00-199-0476
01	18203	A303M	5985-00-199-0476
01	16179	20510-3	5985-00-199-0476
01	93459	3MS-3	---
01	99899	4772-3	5985-01-106-8490
01	16179	20510-3	5985-01-106-8490
01	99899	4771-3	5985-00-115-8130
01	99899	4772-3	5985-00-147-9452
01	99899	4778-3	5985-01-127-3114
01	16179	20640-03	5985-01-127-3114
01	16179	2082-6181-03	5985-01-084-9086
01	18203	A403M	5985-00-443-9175
01	23042	FP-50-3SMA	5985-00-200-9876
01	34078	238-3	5985-01-057-5088
01	34078	205-3	5985-00-152-7923
02	99899	4778-6	5985-00-236-7984
02	16179	20610-6	5985-00-236-7984
02	49956	2888573-2	---
02	16179	20510-6	5985-00-421-1262
02	24930	G30025851	5985-00-025-9294
02	93459	3M-6	5985-00-153-3798
02	93459	3MS-6	5985-01-068-9959
02	16179	2082-4085-06	5985-00-496-3822
02	18203	A380ES	5985-01-068-9959
02	18203	A306M	5985-00-496-3822
02	18203	A406M	5985-01-032-4612
02	34078	238-6	5985-01-056-6390
02	23042	FP-50-6SMA	5985-01-144-8076
02	99899	4771-6	5985-00-115-8131
02	99899	4772-6	5985-00-368-2395

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TABLE III. Substitution information - Continued.

Military part number M3933/14-	Manufacturer's code	Manufacturer's part number	NSN
03	99899	4778-10	5985-00-402-1615
03	16179	20610-10	5985-00-402-1615
03	49956	2888573-3	---
03	93459	3M-10	6625-00-254-5533
03	07397	26-807036-1	5985-00-199-0475
03	16179	20510-10	5985-00-199-0475
03	18203	A310M	5985-00-199-0475
03	23042	FP-50-10SMA	5985-01-144-8076
03	99899	4771-10	5985-00-115-8129
		4772-10	
03	34078	238-10	5985-01-056-6388
03	34078	205-10	5985-01-094-2993
03	18203	A410M	5985-00-428-7360
03	18203	A381ES	5985-01-068-9960
03	16179	2082-6138-10	5985-01-084-9085
			5985-01-093-2390
03	93459	3MS-10	5985-01-068-9960
03	93459	3M-10	5985-01-015-4408
04	99899	4778-20	5985-00-470-9256
04	49956	2888573-4	---
04	93459	3M-20	5985-01-030-7821
04	16179	20510-20	5985-00-426-8860
04	93459	3MS-20	5985-01-070-1155
04	18203	A8120M	5985-01-017-6640
04	18203	A382ES	5985-01-070-1155
04	99899	4772-20	5985-00-341-0589
05	93459	3M-15	4935-01-005-8276
05	99899	4772-15	5985-01-140-8124
05	34078	205-15	5985-01-094-2994
05	18203	A415M	5985-01-032-5452
06	03538	77C712632-1	5985-01-024-1987
06	99899	4778-1	5985-01-024-1987
06	93459	3M-1	---
06	99899	4778-1	5985-01-149-1916
06	23042	FP-50-1SMA	5985-00-431-0354
06	18203	A401M	5985-01-031-6976
06	18203	A301M	5985-00-450-0218
06	16179	2082-6171-01	5985-00-270-4509
06	16179	20510-1	
06	16179	2082-6181-01	5985-01-084-9087
07	03538	77C712632-2	---
07	99899	4778-2	5985-01-091-9615
07	93459	3M-2	5985-01-127-2135
07	34078	238-2	5985-01-058-6389
07	18203	A302M	5985-00-220-2312
07	18203	A402M	5985-01-032-8374
07	16179	20510-2	5985-00-470-2899
08	99899	4778-4	---
08	93459	3M-4	5985-01-127-2136
08	18203	A404M	5985-00-428-7349

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TABLE III. Substitution information - Continued.

Military part number M3933/14-	Manufacturer's code	Manufacturer's part number	NSN
09	99899	4778-5	
09	93459	3M-5	5985-01-120-9265
09	18203	A405M	5985-01-031-5975
09	34078	238-5	5985-01-083-4870
10	93459	3M-7	5985-01-091-8688
10	99899	4778-7	---
11	99899	4778-8	5985-01-036-3461
11	93459	3M-8	5985-01-081-1059
12	99899	4778-9	---
12	34078	238-9	5985-01-083-4238
12	93459	3M-9	5985-01-127-3939
13	99899	4778-30	5985-00-032-3619
13	16179	20510-30 and 5554	5985-00-824-3583
13	97942	128C539H05	5985-00-824-3583
13	93459	3M-30	---
13	99899	4772-30	5985-00-356-7277
13	18203	A430-M	5985-01-031-4582
14	99899	4778-40	---
14	16179	20510-40	5985-00-824-3584
14	97942	128C539H06	5985-00-824-3584
14	99899	4772-40	5985-01-105-8541
14	93459	3M-40	5985-01-093-2591
15	93459	3M-60	5985-01-091-1915
15	81755	C10067-1	---
15	99899	4778-60	5985-00-315-6184
16	93459	3M-50	5985-01-124-2508
17	06401	7910831-00	---
17	34078	205-28	---
18	34078	205-16	5985-01-035-8950
19	99899	4778-14	5985-01-074-3975
20	34078	205-13	5985-01-037-6874
21	99899	4772-12	5985-01-021-5718
21	93459	3M-12	5985-01-139-1004

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TABLE III. Substitution information - Continued.

Military part number M3933/14-	Manufacturer's code	Manufacturer's part number	NSN
22	34078	238-11	5985-01-057-2746
23	93459	3M1.5	5985-01-010-1291
24	34078	238-31	5985-01-083-4239

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - MI
Air Force - 11, 99
DLA - ES

User activities:
Navy - MC, SH, AS
Army - AR

Agent:
DLA - ES

Preparing activity:
Navy - EC

(Project 5985-0983-9)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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